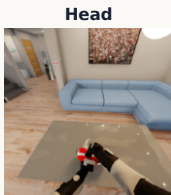


Observation s (chunk start)



Left wrist



Right wrist



Tu... ver

PaliGemma $\pi_{0.5}$
frozen

PaliGemma

gemma_2b · frozen

3 RGB + prompt
→ prefix tokens

prefix KV cache
for the expert

proprio 23-d
(pad → 32) for Q

LoRA actor · one-way sandwich

LoRA rank 32 / α 32 on attn+FFN only · width 1024 · B=0 · lora_scale 0→1

action_in_proj
32 → 1024 frozen

time_mlp (adaRMS)
 $t \rightarrow \text{cond}$ frozen

gemma_300m + LoRA

base weights frozen · LoRA on attn/FFN
prefix KV from PaliGemma

action_out_proj 1024 → 32-d velocity v_t frozen

$\tilde{a} = \text{Euler}(v_{\text{frozen}}, \epsilon)$
10 steps stop-grad

$a = \text{Euler}(v_{\text{live}}, \epsilon)$
same ϵ live grad

$L_{\pi} = \lambda_{\pi}(-\min Q_k) + \beta \|a - \tilde{a}\|^2 + L_{bc}$
 $\beta = 100 \text{ per-dim}$ · $\lambda_{\pi} = 0$ until online
projections frozen so lora_scale=0 is an exact pt12 copy

AE / RL critic for Q only

CFTokenPool

256-d · 4 heads · 2 layers
8 queries → pooled z

CFTokenDecoder

$L_{ro} = \text{MSE}(\hat{p}, \text{sg}(\text{prefix}))$

critic state

$\phi(s) = [\text{sg}(\text{target_pool}), \text{proprio}]$
online: freeze_pool = 1

Q (s, flatten(a))
chunk_start

CFTrunk 1536

input [z, a] 1024+32+736
10 heads $Q = \min_k Q_k$

TD (live)

$y = \text{clip}(r + \gamma(1 - d)Q^-(s', a'), 0, 1)$
 $\gamma = 0.99$ per chunk, not γ^{32}

bootstrap $a'(s')$

EMA LoRA 0.999
actor uses sg(critic)

ice-blue frozen (VLM, expert, projections, target pool, EMA LoRA)

coral trainable (live LoRA, critic; pool only in AE)

dashed stop-grad / auxiliary

solid forward